

★ RS004 Week 14 — English Worksheet

Topic: Vertical Patrol + Death Effect (Apply) · **Course:** Platformer Game · **Time:** about 45 minutes

This week is about thinking through **smarter enemies** and explaining how they work. A `patrol_type` lets one enemy walk left-right, another go up-down, and a third **chase** the player. When the player dies, a little **explosion** of particles bursts out. You will read this code closely and explain the code you wrote in your live lesson.

💬 Words to know: `patrol_type`, **vertical**, **chase**, **particle**, **explosion**.

1 · Predict 🧠

Read each snippet. Write what you think it does.

```
if self.patrol_type == "vertical":
    self.rect.y += self.direction * self.speed
    if self.rect.y > self.start_y + self.range_size:
        self.direction = -1
    if self.rect.y < self.start_y:
        self.direction = 1
```

Which way does this enemy move? When does it turn around?


```
elif self.patrol_type == "chase" and player_rect is not None:
    if self.rect.x < player_rect.x:
        self.rect.x += 1
    elif self.rect.x > player_rect.x:
        self.rect.x -= 1
```

The player is to the *right* of this enemy. Which way does it move?


```
for p in self.particles:  
    p["x"] += p["vx"]  
    p["y"] += p["vy"]  
    p["vy"] += 0.3  
    p["life"] -= 1
```

Each particle has its own speed. Why does `vy` keep growing (`+= 0.3`)?

2 · Pick the Patrol 🕸️

Match each `patrol_type` to its behaviour.

1. "horizontal" 2. "vertical" 3. "chase"

- A. moves up and down between two heights
- B. follows the player when nearby
- C. walks left and right between two x positions

1 → __ 2 → __ 3 → __

3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — A vertical enemy should bounce between two heights. Right now it floats up forever and leaves the screen.

```
if self.patrol_type == "vertical":
    self.rect.y += self.direction * self.speed
```

Hint: there is no turn-around check on the y edges.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A chasing enemy needs the player's position. Right now it crashes with a `NoneType` error.

```
def update(self):
    if self.patrol_type == "chase":
        if self.rect.x < player_rect.x:
            self.rect.x += 1
```

Hint: `player_rect` must be passed in, with a safe default, and used only when it exists.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Dead particles should be removed so the explosion fades. Right now they pile up and slow the game down.

```
for exp in explosions:
    exp.update()
    exp.draw(screen)
```

Hint: when an explosion `is_done()`, take it out of the list. Loop over a copy while removing.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Explain the Code

Here is a working enemy update that handles all three patrol types and takes the player's position safely.

```
def update(self, player_rect=None):
    if self.patrol_type == "horizontal":
        self.rect.x += self.direction * self.speed
        if self.rect.x > self.start_x + self.range_size:
            self.direction = -1
        if self.rect.x < self.start_x:
            self.direction = 1

    elif self.patrol_type == "vertical":
        self.rect.y += self.direction * self.speed
        if self.rect.y > self.start_y + self.range_size:
            self.direction = -1
        if self.rect.y < self.start_y:
            self.direction = 1

    elif self.patrol_type == "chase" and player_rect is not None:
        if self.rect.x < player_rect.x:
            self.rect.x += 1
        elif self.rect.x > player_rect.x:
            self.rect.x -= 1
```

What does `self.direction = -1` do, and when does it happen?

Why does the `vertical` block change `self.rect.y` instead of `self.rect.x`?

Why does the `chase` block check `player_rect is not None` before using it?

What would break if you deleted the `start_x` and `start_y` checks?

Tell the story of one chase frame: the player is to the left, so what happens line by line?

5 · Explain Your Lesson Code 🎥

In your live lesson today you wrote enemy and explosion code. Make a short phone video that explains **the code you wrote**. You may show it running. Try to use these words: **patrol_type**, **vertical**, **chase**, **particle**, **explosion**.

1. Point to where `patrol_type` decides how each enemy moves.
2. Read your chase code out loud and explain how it picks a direction.
3. Show the death **explosion** and explain how the **particle** loop works.
4. Say one part that was tricky and how you understood it.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.